Rheinisch-Westfälische Technische Hochschule Aachen
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Phase Review: 1st Sprint of the Implementation of "Conformance Predictions"

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Contents

1 Summary of Sprint 1 **3**

2 Codebase Structure and CLI Methodology **3**

3 Requirements Coverage Status - Overview per Module **3**
3.1 Running High-level Requirements 3
3.2 IO 3
3.3 Conformance 4
3.4 Feature Extraction 4
3.5 Conformance 4
3.6 Model Training 4
3.7 Evaluation 4
3.8 Config & Pipeline 4
3.9 Project Infrastructure & Testing 4

4 Plan for next sprint and remaining requirements **5**

5 User Guide **6**

6 Team settings and Workflow **6**
6.1 Changes in Responsibilities 6
6.2 Workflow Improvements for the next sprint 7

7 Exemplary Visualizations and Outputs **7**

8 Sprint review **7**
8.1 Richard Schuppener 7
8.2 Cem Colak 7

1 Summary of Sprint 1

Achievements of Sprint 1 (based on GitLab work-items) from high level perspective, short and no technical details

2 Codebase Structure and CLI Methodology

We designed the structure of the codebase in the following philosophy: modules for each step of the process that encapsulate all raw functionalities, pipelines that combine the modules functionalities bundled in a pipeline module and a descriptive, concise and powerful CLI that guides the user through the whole process and cleverly calls the pipelines with the data and options selected by the user. We decomposed the modules into the following categories:

1. IO: Handle inputs and outputs, such as reading of event logs, normative models, visualization of process models etc.
2. Feature Extraction:
3. Model Training:
4. Evaluation:
5. Config:
6. Pipeline:
7. Main / CLI:

3 Requirements Coverage Status - Overview per Module

3.1 Running High-level Requirements

3.2 IO

Covered Planned in Sprint 2

3.3 Conformance

Covered

Planned in Sprint 2

3.4 Feature Extraction

Covered

Planned in Sprint 2

3.5 Conformance

Covered

Planned in Sprint 2

3.6 Model Training

Covered

Planned in Sprint 2

3.7 Evaluation

Covered

Planned in Sprint 2

3.8 Config & Pipeline

Covered

Planned in Sprint 2

3.9 Project Infrastructure & Testing

Covered

Planned in Sprint 2

LL-01.1 LL-01.2 LL-03.1 HL-03 Partially: Missing conformance deviations prediction as target and case attributes as features HL-04: Reporting errors, preprocessing failures, and training failures in CLI HL-05: loading and preprocessing event log LL-05.1: Parsing of CSV event logs supported for CSV and XES LL-05-2: Partially: petri net loading works LL-05.3: Representation of event log with petri net, BPMN & Process Tree by missing provision of a normative model HL-06: Partially: alignment based diagnostics in preprocessing covered LL-06.1: Raw alignment dictionaries for events against the normative model LL-06.2: Cumulative alignment costs for each prefix HL-07: Partially: log-based and conformance-based features implemented in preprocessing LL-07.3: Control-flow extracted features for each event log case HL-08: Partially: Supervised prediction pipeline implemented, but only with decision tree LL-08.1: Standardization of features through standard scaler in machine learning module HL-09: Baseline and conformance aware model implemented HL-10: End-to-End experiment runner implemented HL-12:

Conformance: HL-13: cached conformance analysis results for repeated use during feature generation HL-18: Used PM4Py for conformance checking and feature extraction and scikit-learn for model training alongside pandas, numpy and matplotlib HL-19: We decided to put conformance and feature extraction into feature extraction module

Evaluation H-15: Partially: temporal train/test split for evaluation. Left out validation split and time order based splits HL-16: Additional evaluation metrics for predictive performance (e.g., precision, recall, F1-score, ROC-AUC) HL-20: Top-Level requirements HL-12: Usage of Matplotlib HL-19: Modular structure of the project, usage of git for version control HL-20: README file

4 Plan for next sprint and remaining requirements

HL-02: inclusion of case attributes

HL-01, HL-03, conformance deviations prediction as target variable HL-03 Conformance deviations as target variable LL-05.2: Loading of BPMN and process tree models HL-06: maybe distance to completion indicators LL-07.1: Truncating event logs to prefixes of varying lengths LL-07.2: Formatting numerical features for remaining time predictions LL-08.2: Fitted predictive model capable of mapping the feature matrix to the deviation risk target HL-08: Case final alignment costs HL-11: Automated testing via a

single command LL-11.1: Isolated and predictable testing environments containing synthetic process data LL-11.2: Automated assertion of correctness for the core conformance logic. LL-11.3: Automated verification of the prefix generation to prevent data leakage

Feature extraction: LL-13.1: Disk-based persistence of intermediate alignment results

Evaluation: LL-13.1: Identification and usage of prefix length calculation to avoid redundancies HL-14: Measuring and reporting runtime and resource requirements of conformance checking and feature generation LL-14.1: High precision metrics on the execution duration of critical script segments LL-14.2: Tracking of peak memory footprint during data preprocessing and alignment LL-14.3: Start and End recording for execution of conformance checking and feature generation per individual data batch LL-14.4: Aggregation and reporting of runtime and resource usage statistics in the console and final evaluation report HL-15: Left out validation split and time order based splits HL-16:conformance checking (e.g., fitness, precision, generalization, simplicity) for the normative model LL-16.2: MAE/RMSE for regression tasks as evaluation metrics for predictive performance HL-17: Analysis and reporting for prefix length based performance evaluation

Requirements that are to be implemented (that have not been implemented in Sprint 1) Functionalities that we did not specify as requirements but would still be nice to have

5 User Guide

Refer to existing user guide and that we will maintain the user guide throughout the sprints.

6 Team settings and Workflow

6.1 Changes in Responsibilities

Mention that we will change the responsibilities for the next sprint to enhance the learning experience and gain more insights in both modules

6.2 Workflow Improvements for the next sprint

Mention our aim towards a more granular breakdown of the work items: specific and better traceable requirement coverage. Mention that we will maintain the breakdown of the requirements per module listed here in the review document and update it after each sprint to keep track of the coverage status.

7 Exemplary Visualizations and Outputs

Exemplary visualizations (from IO, Evaluation module) and numerical results e.g. accuracy, class accuracy (from Evaluation module)

8 Sprint review

8.1 Richard Schuppener

8.2 Cem Colak